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Notifications 

3. General solution to inhomogeneous systems

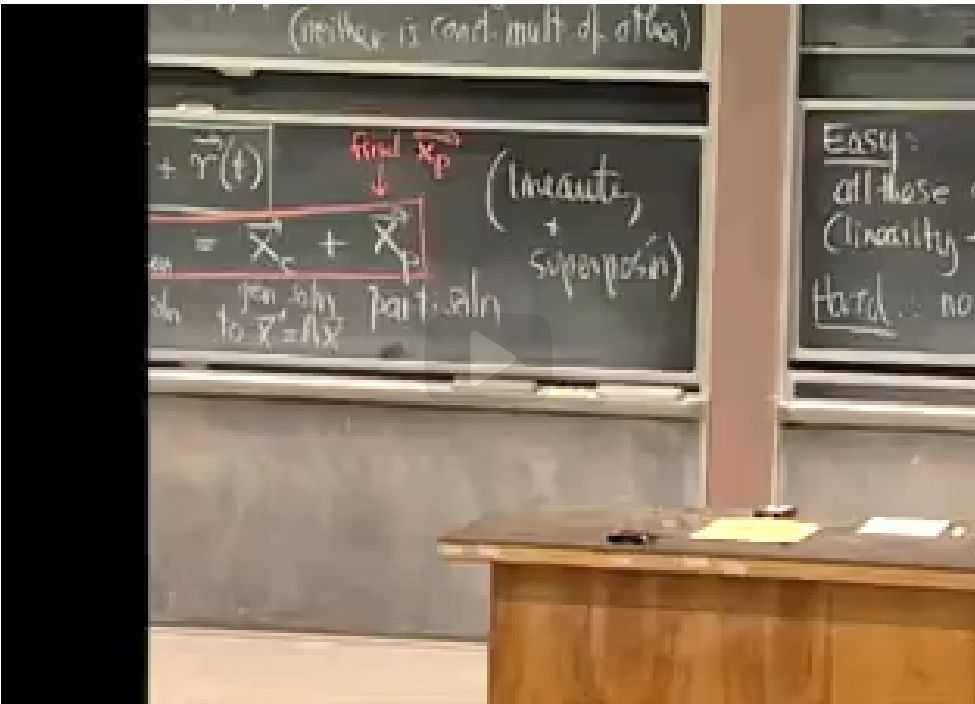
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Introduction to inhomogeneous system



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right you are--
a particular solution.
That's the thing you couldn't
proof your last exam,
so maybe I'll give it to you on
this exam to prove,
and maybe I won't.
One of those two things.
But the principle is the same,
and it's
proved exactly the same way.
It's just linearity and
superposition.
The linearity of the system,
original system,
and the superposition principle.
So the essence is that to solve
this inhomogeneous system,
what
we have to do is find a particular

Recall that an **inhomogeneous** first order $n \times n$ linear system of ODEs is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{r}(t) \tag{4.116}$$

where $\mathbf{A}$ is an $n \times n$ matrix, $\mathbf{r}(t)$ is a vector in n -dimensional space, and they both depend only on the independent variable t .

The general solution to such an inhomogeneous system is

$$\mathbf{x}(t) = \underbrace{\mathbf{x}_h(t)}_{\text{homogeneous}} + \underbrace{\mathbf{x}_p(t)}_{\text{particular}}, \tag{4.117}$$

where $\mathbf{x}_h$ is the general solution to the associated homogeneous system:

$$\dot{\mathbf{x}}_h = \mathbf{A}\mathbf{x}_h, \tag{4.118}$$

and $\mathbf{x}_p$ is one particular solution satisfying the full inhomogeneous equation:

$$\dot{\mathbf{x}}_p = \mathbf{A}\mathbf{x}_p + \mathbf{r}. \tag{4.119}$$

This is due to the linearity of the system and the superposition principle.

As before in this course, we will restrict ourselves to the case when $\mathbf{A}$ is **constant**.

Review: companion system

2/2 points (graded)

For the second order single ODE

$$\ddot{x} + x = \tan(t), \tag{4.120}$$

find the companion system with new variable $y = x$:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{r} \quad \text{where } \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \tag{4.121}$$

by entering the companion matrix $\mathbf{A}$ and the vector $\mathbf{r}$ below

(Enter `[[a,b],[c,d]]` for the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. That is, use commas to separate entries within the same row and then enclose the entire row in **square brackets**. Then use **commas** to separate rows, and **square bracket** the entire matrix. Similarly for larger matrices.)

$\mathbf{A} =$

✓

 Answer: `[[0,1],[-1,0]]`

(Enter a single vector as your answer. Enter `[a , c]` for the vector $\begin{pmatrix} a \\ c \end{pmatrix}$. That is, use **comma** to separate the entries, and **square bracket** the entire vector.)

$\mathbf{r} =$

✓

 Answer: `[0, tan(t)]`

Solution:

Let $y = \dot{x}$, then the companion system is

$$\dot{x} = y \tag{4.122}$$

$$\dot{y} = -x + \tan(t). \tag{4.123}$$

In matrix form, this is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{r} \quad \text{where } \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \tag{4.124}$$

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \tag{4.125}$$

$$\mathbf{r}(t) = \begin{pmatrix} 0 \\ \tan(t) \end{pmatrix}. \tag{4.126}$$

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You have used 2 of 3 attempts

i Answers are displayed within the problem

3. General solution to inhomogeneous systems

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